Governing equations and their classification

Prognostic Modelling (pea700) · Lecture 2

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Recap of lecture 1

- A prognostic model steps the state forward with equations containing $\partial/\partial t$
- Discretization: derivatives become difference quotients
- Forward Euler for $\mathrm{d}u/\mathrm{d}t = -ku$: first order, unstable for $k\Delta t > 2$
- Floating point: 16 digits, loss of significance, conditioning

Today

- The language: gradient, divergence, curl, Laplacian
- Conservation laws: from a balance to a PDE
- Classification: ODE/PDE, order, linearity, elliptic/parabolic/hyperbolic
- Prototype equations: diffusion, advection, waves, Poisson
- The equations of atmosphere, ocean, groundwater and seismics, and their types
- Initial and boundary conditions, well-posedness

> Classify before you compute: the type of an equation decides which numerical methods work.

Vector calculus

Notation: derivatives and vectors

Derivatives

- one variable, $u(t)$: ordinary derivative $\dfrac{\mathrm{d}u}{\mathrm{d}t}$
- several variables, $u(x,t)$: partial derivatives $\dfrac{\partial u}{\partial t}$, $\dfrac{\partial u}{\partial x}$, the other variables held fixed
- order n: $\dfrac{\mathrm{d}^n u}{\mathrm{d}t^n}$, $\dfrac{\partial^n u}{\partial x^n}$
- short forms: u_t, u_x, u_{xx}

Vectors

$$\vec{v} = v_1\vec{i} + v_2\vec{j} + v_3\vec{k}$$

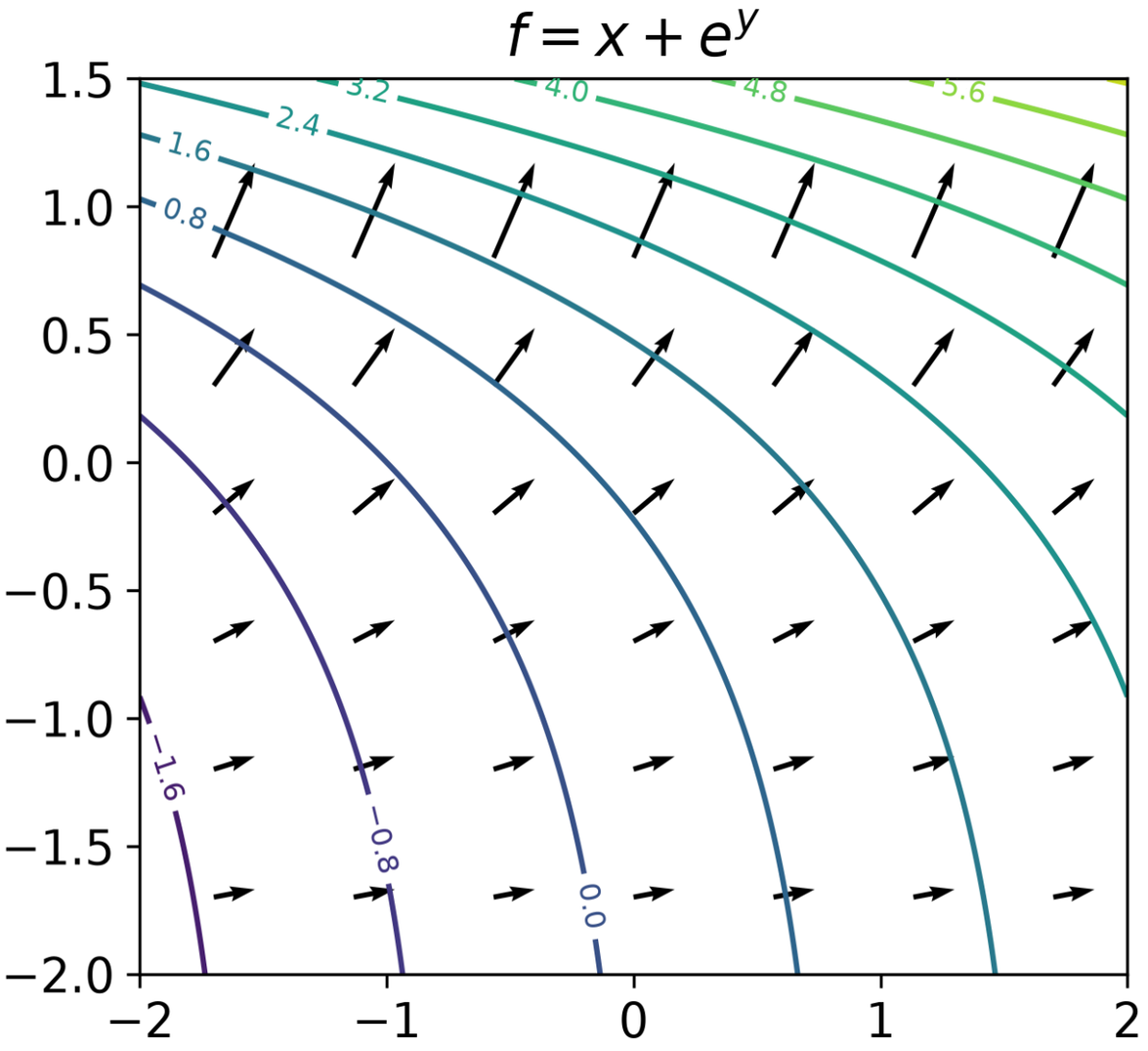
- $\vec{i}$, $\vec{j}$, $\vec{k}$: unit vectors along the axes
- magnitude $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$
- unit vector in the direction of $\vec{v}$: $\vec{v}/\|\vec{v}\|$

The gradient

$$\nabla f = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j} + \frac{\partial f}{\partial z}\vec{k}$$

- points to the **steepest increase**
- perpendicular to the contours
- magnitude: rate of change, large where contours are dense

Gradients drive fluxes: heat, groundwater, wind.



The divergence: two definitions

Outflow per volume, geometric:

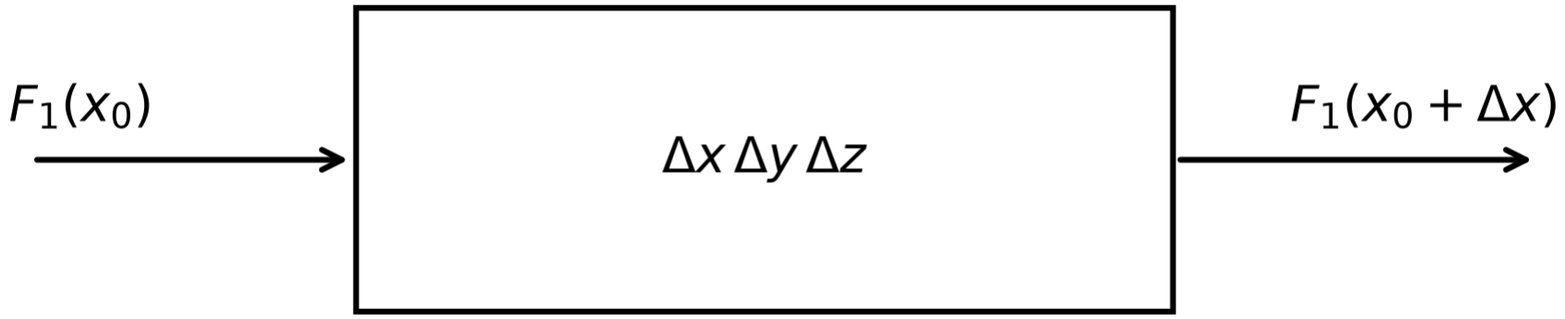
$$\nabla \cdot \vec{F} = \lim_{V \to 0} \frac{1}{V} \oint_S \vec{F} \cdot \mathrm{d}\vec{A}$$

Cartesian:

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Why are they the same?

Why both definitions agree



Flux through the two faces normal to x (outward normals $-\vec{i}$ and $+\vec{i}$):

$$\left[F_1(x_0+\Delta x) - F_1(x_0)\right]\Delta y\,\Delta z = \frac{F_1(x_0+\Delta x) - F_1(x_0)}{\Delta x}\,\Delta x\,\Delta y\,\Delta z \approx \frac{\partial F_1}{\partial x}\,V$$

Same for y and z; divide by V. **This is exactly what the finite volume method does.**

Two results we use constantly

Product rule

$$\nabla \cdot (g\vec{F}) = g\,\nabla \cdot \vec{F} + \vec{F} \cdot \nabla g$$

Divergence theorem (Gauss)

$$\int_V \nabla \cdot \vec{F}\,\mathrm{d}V = \oint_S \vec{F} \cdot \mathrm{d}\vec{A}$$

What flows out through the surface equals the sum of the sources inside.

Curl

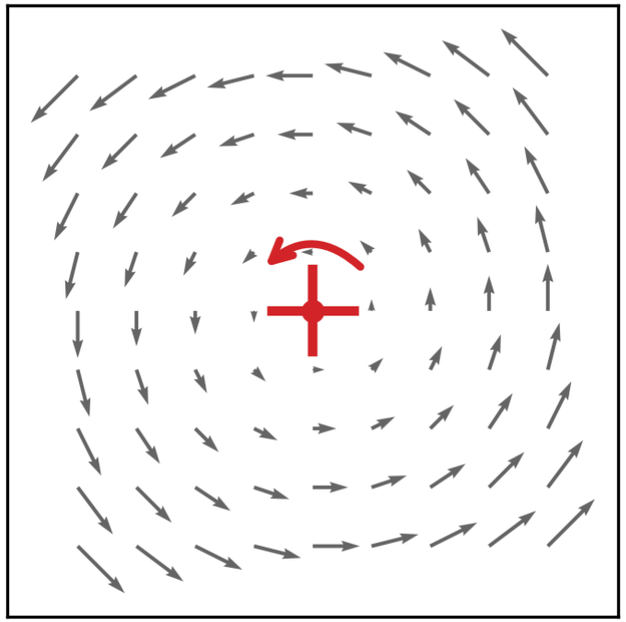
Curl $\nabla \times \vec{F}$: local rotation. Vertical component for a horizontal wind is the relative vorticity

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

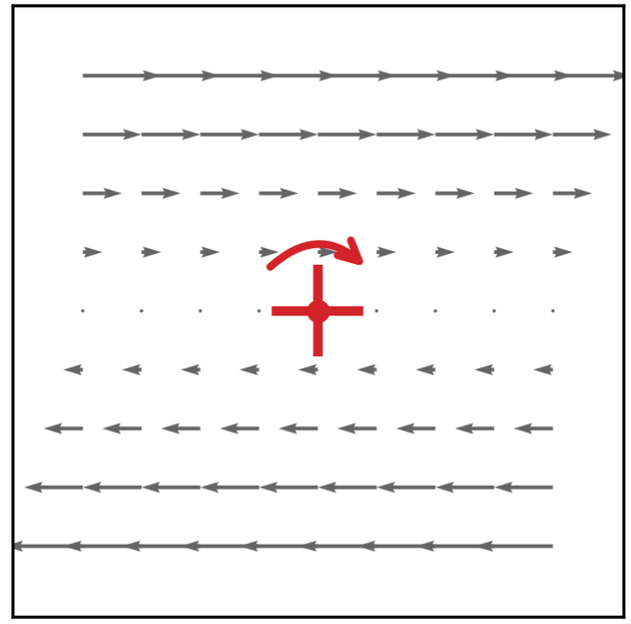
A paddle wheel in the flow spins with $\zeta/2$, anticlockwise for $\zeta > 0$.

Shear has vorticity, even with **straight streamlines**.

rotation: $\zeta = 2$



shear: $\zeta = -1$



Laplacian

Laplacian: divergence of the gradient

$$\nabla^2 f = \nabla \cdot (\nabla f) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Compares f with the average of its neighbourhood: $\nabla^2 f < 0$ at a maximum. It appears in every equation that **smooths**.

Conservation laws

A balance over a control volume

u: density of the conserved quantity, $\vec{F}$: its flux, s: sources

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_V u\,\mathrm{d}V = -\oint_S \vec{F}\cdot\mathrm{d}\vec{A} + \int_V s\,\mathrm{d}V$$

1. Fixed volume: $\frac{\mathrm{d}}{\mathrm{d}t}\int_V u\,\mathrm{d}V = \int_V \frac{\partial u}{\partial t}\,\mathrm{d}V$
2. Gauss: $\oint_S \vec{F}\cdot\mathrm{d}\vec{A} = \int_V \nabla\cdot\vec{F}\,\mathrm{d}V$
3. Collect: $\int_V \left(\frac{\partial u}{\partial t} + \nabla\cdot\vec{F} - s\right)\mathrm{d}V = 0$ **for every** V

$$\boxed{\frac{\partial u}{\partial t} + \nabla\cdot\vec{F} = s}$$

Closing the equation: flux laws

Advective flux: carried by the flow

$$\vec{F} = \vec{v}\, u$$

Diffusive flux: down the gradient

$$\vec{F} = -D \nabla u$$

Law	Flux	Driven by
Fourier	heat	temperature gradient
Fick	dissolved substance	concentration gradient
Darcy	groundwater	gradient of hydraulic head

Where did the minus sign go?

Insert $\vec{F} = \vec{v}\,u - D\,\nabla u$ into $\partial u/\partial t + \nabla \cdot \vec{F} = s$:

$$\frac{\partial u}{\partial t} + \nabla \cdot (\vec{v}\,u) \underbrace{-\nabla \cdot (D\,\nabla u)}_{\text{from } -D\nabla u} = s \quad \Longrightarrow \quad \boxed{\frac{\partial u}{\partial t} + \nabla \cdot (\vec{v}\,u) = \nabla \cdot (D\,\nabla u) + s}$$

Two minus signs cancel: outward normal (outflow reduces u) and flux law (down the gradient).

Sign errors run, but give wrong results. Conventions differ:

normal inward or outward · z up or down · $-\nabla^2 u = f$ or $\nabla^2 u = f$ · source or sink

- write the convention down, in the derivation and in the code
- check the physics: at a maximum $\nabla^2 u < 0$, so u must decrease
- test: a Gaussian must spread, $\int u \,\mathrm{d}V$ must stay constant

Flux form and advective form

$$\nabla \cdot (\vec{v} u) = \vec{v} \cdot \nabla u + u \, \nabla \cdot \vec{v}$$

For incompressible flow, $\nabla \cdot \vec{v} = 0$:

$$\underbrace{\frac{\partial u}{\partial t} + \nabla \cdot (\vec{v} u) = \ldots}_{\text{flux form}} \qquad \Longleftrightarrow \qquad \underbrace{\frac{\partial u}{\partial t} + \vec{v} \cdot \nabla u = \ldots}_{\text{advective form}}$$

> Mathematically equivalent, numerically not: only a discretization of the **flux form** conserves the total amount exactly. What leaves one cell enters its neighbour.

Classification

ODE, PDE, systems, order

- **ODE**: one independent variable. **PDE**: several, partial derivatives
- **System**: several unknowns, several equations (predator–prey; atmosphere and ocean models)
- **Order**: highest derivative; decides the number of initial and boundary conditions

Every ODE of order n is a system of n first-order ODEs. Newton:

$$m\frac{\mathrm{d}^2u}{\mathrm{d}t^2} = F \qquad \Longrightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}t} = v, \quad \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F}{m}$$

ODE software is written for first-order systems.

Linear or nonlinear?

Think first: order and linearity?

Equation	Order	Linear?
$y' - 2xy = e^x$	1	yes
$y'' + 4y' + 4y = 0$	2	yes, homogeneous
$y' = x/y$	1	no
$y'' + \sin y = x^2$	2	no
$u_t + c(x)\, u_x = 0$	1	**yes**, despite variable $c(x)$
$u_t + u\, u_x = 0$	1	**no**: the velocity advects itself

Linear: superposition, wave-by-wave analysis, linear algebraic systems. Nonlinear: iteration (Newton).

Elliptic, parabolic, hyperbolic

$$a\,u_{xx} + 2b\,u_{xy} + c\,u_{yy} + \text{lower-order terms} = f$$

Type	Condition	Prototype	Eigenvalues of $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$
Elliptic	$b^2 - ac < 0$	Poisson $u_{xx} + u_{yy} = f$	same sign
Parabolic	$b^2 - ac = 0$	diffusion $u_t = Du_{xx}$	one is zero
Hyperbolic	$b^2 - ac > 0$	wave $u_{tt} = c^2 u_{xx}$	opposite signs

Check: diffusion $u_t - Du_{xx} = 0$ in (x, t)?

$a = -D, b = c = 0$: $b^2 - ac = 0$, parabolic. Wave: $a = -c_0^2, c = 1$: $b^2 - ac = c_0^2 > 0$, hyperbolic.

Prototype equations

Diffusion, advection, waves

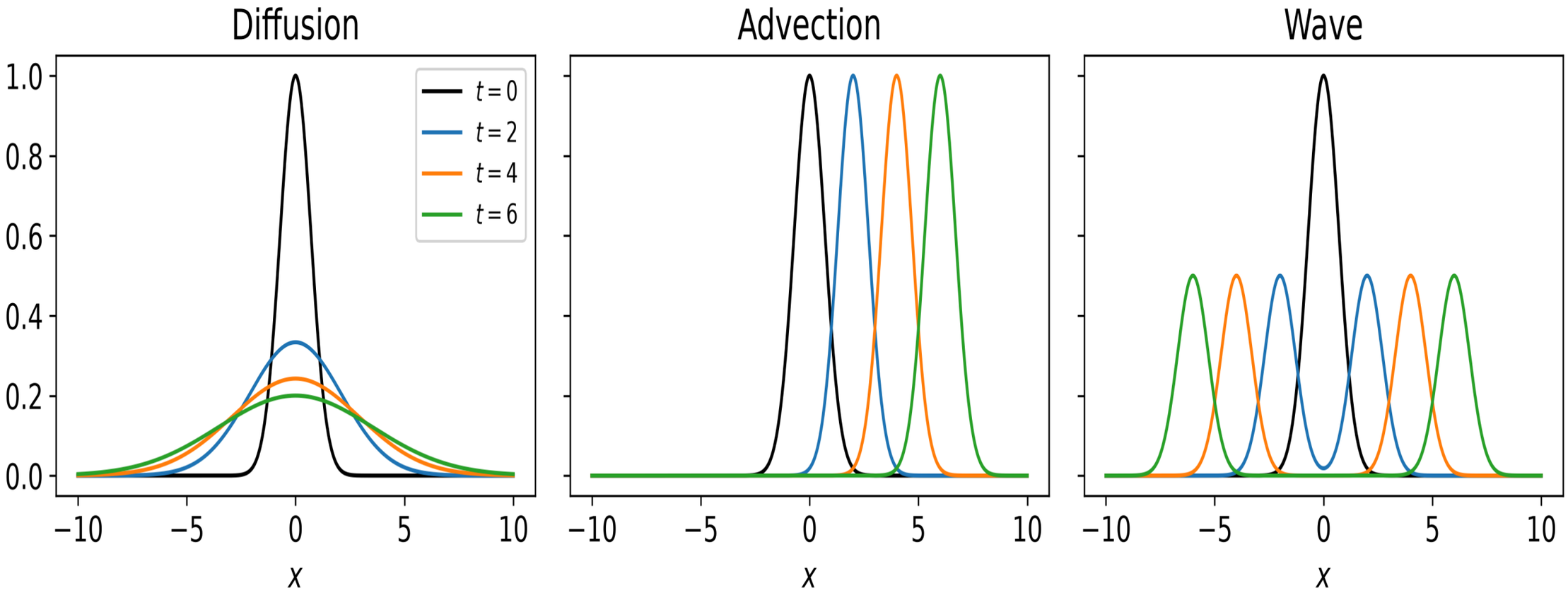
$$\underbrace{\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2}}_{\text{diffusion: parabolic}} \qquad \underbrace{\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0}_{\text{advection: hyperbolic}} \qquad \underbrace{\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}}_{\text{wave: hyperbolic}}$$

Exact solutions for $u(x,0) = f(x)$:

- **Advection**: $u = f(x - ct)$. Check with $\xi = x - ct$: $u_t = -cf'(\xi)$, $u_x = f'(\xi)$
- **Wave**: $u = \frac{1}{2}[f(x - ct) + f(x + ct)]$ (zero initial velocity)
- **Diffusion**, Gaussian $f = e^{-x^2/w^2}$:

$$u = \frac{w}{\sqrt{w^2 + 4Dt}} \exp\left(-\frac{x^2}{w^2 + 4Dt}\right)$$

Three behaviours



Diffusion **smooths** and has no direction. Advection and waves **transport** with finite speed and keep the shape. Numerical methods must not smear or distort a travelling signal: the harder requirement.

Steady state: the Poisson equation

$\partial u/\partial t = 0$ in the diffusion equation with a source; elliptic:

$$-\nabla^2 u = f \qquad (\text{Laplace equation for } f = 0)$$

Equilibria: steady temperature and groundwater fields, gravitational potential, pressure in an incompressible fluid.

Why the type matters

	Elliptic	**Parabolic**	**Hyperbolic**
Describes	equilibrium	smoothing	propagation
A disturbance is felt	everywhere at once	everywhere, damped	only where the wave has arrived
Solution is	smoother than the data	smoother than the initial state	as rough as the initial state
Needs	BCs on the whole boundary	IC and BCs	ICs; BCs where information enters
Numerical task	one large linear system	time stepping, often implicit	time stepping, respect the speed

The equations of this lecture

Shallow water equations

$$\frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} = 0, \qquad \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + g\frac{\partial h}{\partial x} = 0$$

Linearize about rest, $h = H + \eta$, small η and u:

$$\frac{\partial \eta}{\partial t} + H\frac{\partial u}{\partial x} = 0, \qquad \frac{\partial u}{\partial t} + g\frac{\partial \eta}{\partial x} = 0$$

∂_t of the first, ∂_x of the second, eliminate u:

$$\frac{\partial^2 \eta}{\partial t^2} = gH\frac{\partial^2 \eta}{\partial x^2} \qquad \Longrightarrow \qquad \text{gravity waves with } c = \sqrt{gH}$$

Ocean depth 4 km: $c \approx 200$ m/s, a tsunami. A **hyperbolic system**; with rotation the test bed for weather and climate model numerics.

First-order systems and characteristics

Advection $u_t + cu_x = 0$: information travels along $x - ct = \text{const}$ (**characteristics**): hyperbolic.

$$\frac{\partial \vec{q}}{\partial t} + B\frac{\partial \vec{q}}{\partial x} = 0 \qquad \text{hyperbolic if } B \text{ has real eigenvalues}$$

(and a full set of eigenvectors): then it splits into advection equations, one per eigenvalue.

Linear shallow water, $\vec{q} = (\eta, u)^T$:

$$B = \begin{pmatrix} 0 & H \\ g & 0 \end{pmatrix}, \qquad \det(B - \lambda I) = \lambda^2 - gH = 0 \;\Rightarrow\; \lambda = \pm\sqrt{gH}$$

The eigenvalues are the wave speeds. Full equations: $u \pm \sqrt{gh}$.

Real models mix types: advection–diffusion is formally parabolic but behaves hyperbolically when advection dominates.

Atmosphere, groundwater, soil

Primitive equations: momentum, mass, energy of a stratified fluid on the rotating sphere; each an advection equation with sources (pressure gradient, Coriolis, heating, phase changes): dominated by hyperbolic behaviour.

Groundwater: mass + Darcy, a diffusion equation (parabolic; elliptic in steady state) with K varying over orders of magnitude

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (K \nabla h) + q$$

Richards equation: unsaturated soil, coefficients depend on the solution: **nonlinear** and parabolic

$$\frac{\partial \theta(\psi)}{\partial t} = \nabla \cdot [K(\psi)\, \nabla(\psi + z)]$$

Seismic waves

Momentum + Hooke's law for a shear wave (velocity v, stress σ):

$$\rho \frac{\partial v}{\partial t} = \frac{\partial \sigma}{\partial x}, \qquad \frac{\partial \sigma}{\partial t} = \mu \frac{\partial v}{\partial x}$$

∂_t of the first, insert the second:

$$\rho \frac{\partial^2 v}{\partial t^2} = \frac{\partial}{\partial x} \frac{\partial \sigma}{\partial t} = \mu \frac{\partial^2 v}{\partial x^2} \qquad \Longrightarrow \qquad c = \sqrt{\mu / \rho}$$

> Same structure as shallow water: two first-order equations for two variables that drive each other.

Conditions and well-posedness

Initial and boundary conditions

Initial conditions: as many as the order in time (diffusion: $u(x, 0)$; wave: u and u_t)

Type	Condition	Meaning	Example
Dirichlet	$u = g$	value prescribed	fixed temperature, lake level at an aquifer
Neumann	$\partial u/\partial n = g$	flux prescribed	insulated wall, impermeable layer ($g = 0$)
Robin	$\partial u/\partial n + \alpha u = g$	flux depends on value	heat exchange with the air

Periodic: $u(0) = u(L)$, exact on a latitude circle. **Open boundaries** of limited-area models: waves must leave without reflection.

Well-posed problems (Hadamard)

1. a solution **exists**
2. it is **unique**
3. it depends **continuously on the data**

- **Too few conditions**: Poisson with Neumann everywhere. $u + C$ is also a solution; a solution exists only if fluxes balance sources. Discrete: singular matrix.
- **Wrong direction of time**: diffusion backward in time amplifies small-scale errors without bound. Inverse problems in geophysics, data assimilation in meteorology.

> Well-posed is not the same as predictable: the atmosphere is well-posed, yet forecasts lose skill after about two weeks.

Summary

- Gradient, divergence, curl, Laplacian: the divergence is the net outflow per volume.
- Conservation law: $\partial u/\partial t + \nabla \cdot \vec{F} = s$, closed by advective and diffusive fluxes.
- Classify first: ODE/PDE, order, linearity, elliptic/parabolic/hyperbolic.
- Prototypes: diffusion smooths; advection and waves propagate with finite speed; Poisson describes equilibria.
- Real models (atmosphere, ocean, groundwater, seismics) combine the prototypes and are classified the same way.
- Well-posed: right number of conditions, continuous dependence on the data.

Lab and next lecture

- **Lab**: start Exercise 2 (gradient, Laplacian, divergence, conservation law, classification), due date
- **Reading**: script, unit 2
- **Next lecture**: the numerical toolbox: finite differences, quadrature, root finding